```
FROM alpine:${ALPINEVER}
LABEL "com.richnusgeeks.vendor"="richnusgeeks"
LABEL "com.richnusgeeks.category"="utility"
LABEL version="1.19.3"
LABEL description="docker based testing driver for container
structure test"

ARG CSTVER=1.19.3

SHELL ["/bin/ash", "-x", "-o", "pipefail", "-c"]
WORKDIR /tmp
RUN apk add --no-cache --virtual .dwnld-deps ca-certificates
curl \
  && apk add --no-cache bash \
  && if [[ $(uname -m) == 'aarch64' ]];then CSTPLT=arm64;else
CSTPLT=amd64;fi \
  && curl -sSL -o /usr/local/bin/cstest "https://github.
com/GoogleContainerTools/container-structure-test/releases/
download/v${CSTVER}/container-structure-test-$(uname|tr A-Z
a-z)-${CSTPLT}" \
  && chmod +x /usr/local/bin/cstest \
  && mkdir -p /etc/cstest \
  && if [[ "${CSTPLT}" == 'arm64' ]];then CSTPLT=aarch64;else
CSTPLT='x86_64';fi \
  && curl -sSL -o /tmp/docker.tgz "https://download.docker.
com/linux/static/stable/${CSTPLT}/$(curl -sSkL https://
download.docker.com/linux/static/stable/${CSTPLT}/|grep
'^ *<a'|grep docker|grep -v rootless|awk -F '\"' '{print
$2}'|sort -nr|head -1)" \
  && tar zxvf docker.tgz

WORKDIR /etc/cstest

ENTRYPOINT ["cstest"]
CMD ["-h"]

--- End of Dockerfile_CSTestPrblm ---
```

Now let's create the test cases file shown below to feed to the CST:

```
config.yaml

schemaVersion: 2.0.0

commandTests:
  - name: "dump cstest version"
    command: "/usr/local/bin/cstest"
    args: ["version"]
    expectedOutput: ["1.1[6-9].[0-9]"]

fileExistenceTests:
  - name: "downloaded docker.tgz absence in /tmp"
    path: "/tmp/docker.tgz"
    shouldExist: false

  - name: "extracted docker folder absence in /tmp"
    path: "/tmp/docker"
    shouldExist: false

  - name: "curl absence in /usr/bin"
    path: "/usr/bin/curl"
    shouldExist: false

  - name: "ca-certificates absence in /etc"
    path: "/etc/ca-certificates"
    shouldExist: false

  - name: "cstest absence in /sbin"
    path: "/sbin/container-structure-test"
    shouldExist: false

  - name: "cstest absence in /bin"
    path: "/bin/container-structure-test"
    shouldExist: false

metadataTest:
  labels:
    - key: 'com.richnusgeeks.vendor'
      value: 'richnusgeeks'
  exposedPorts: []
  volumes: []
  entrypoint: ["cstest"]
  cmd: ["-h"]
  workdir: "/etc/cstest"
```

Finally, let's trigger a static test run to verify the structure of the already created Docker image through a command in your terminal:

```
docker run -it --rm -v ${PWD}:/etc/cstest:ro -v /
var/run/docker.sock:/var/run/docker.sock:ro ghcr.io/
googlecontainertools/container-structure-test:1.19.3 test -c
/etc/cstest/config.yaml -i cstestprblm:1.19.3
```

The output of the command is shown in Figure 1, clearly indicating which test cases have passed or failed against our desired structure of the Docker image.

Now, let's decipher everything line by line. The *config.yaml* is a CST configuration file setting baseline test cases to vet a container image against it. We verify that the CST command packed in the image runs properly and is of a particular version that matches a regular expression under the *commandTests* section. Command tests ensure that certain